



SEMESTER END EXAMINATIONS - AUGUST 2024

Program : **B.E. - Civil Engineering**

Semester : **IV**

Course Name : **Analysis of Determinate and Indeterminate Structures**

Max. Marks : **100**

Course Code : **CV44**

Duration : **3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. Analyze the plane truss shown in Fig. 1 by using method of joints.

CO1 (20)

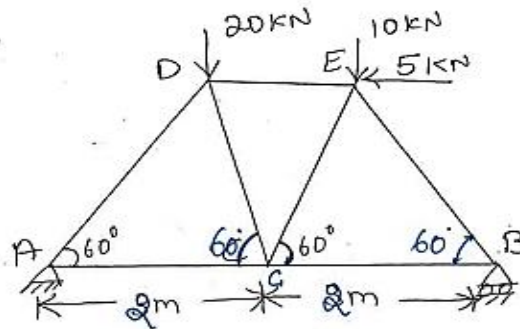


Fig. 1

2. a) Determine the forces in members AB, AD and DE using method of sections. shown in Fig.2(a).

CO1 (10)

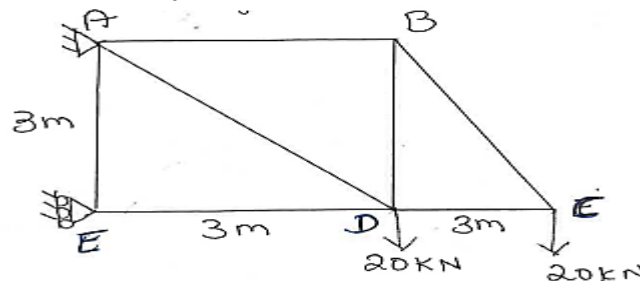


Fig.2(a)

- b) Determine the vertical deflection of joint C for the pin jointed truss shown in Fig.2(b) by using unit load method. The cross sectional areas of the members are shown on the respective members. Take $E = 200\text{kN/mm}^2$.

CO1 (10)

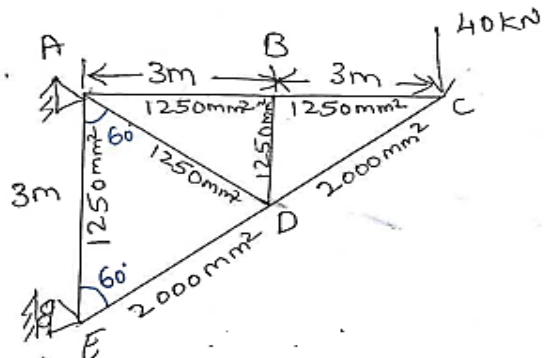


Fig.2(b)

UNIT - II

3. A symmetrical three hinged parabolic arch of span 23m and rise of 3m carries a vertical load of 15kN at 6m from left support. Determine the Reactions, Normal thrust, radial shear and Bending moment at 4m and 6m from right support. CO2 (20)
4. Multiple point loads of 110,125,90 and 180kN with spacing of 3m each crossing over girder of span 30m from left to right with 110kN leading. Determine the reactions, maximum shear force at section 12m from left and also find the maximum BM. CO2 (20)

UNIT - III

5. Analyze the continuous beam shown in Fig. 5 by using slope deflection method and draw BMD. CO3 (20)

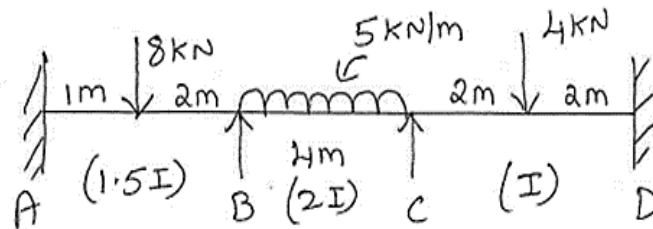


Fig. 5

6. a) Analyze the propped cantilever beam shown in Fig.6(a) and draw BMD. CO3 (10)

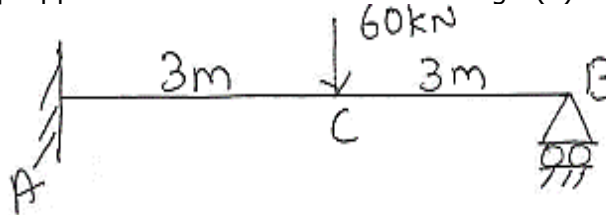


Fig.6(a)

- b) Analyze the Fixed beam shown in Fig.6(b) and draw BMD. CO3 (10)

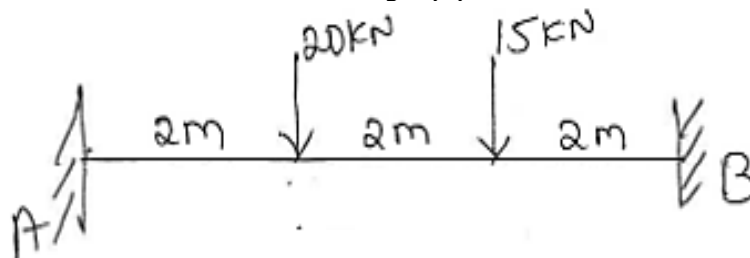


Fig.6(b)

UNIT- IV

7. Analyze the continuous beam as shown in Fig.7 using Moment Distribution method and also Draw BMD. CO4 (20)

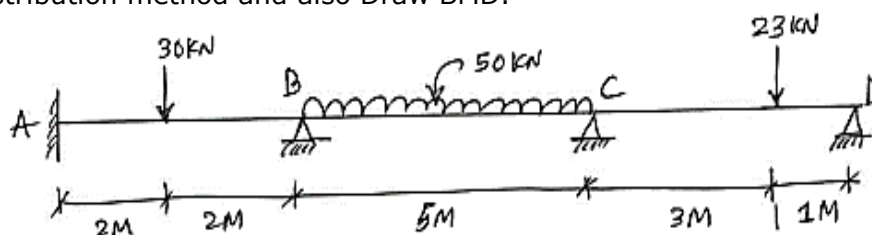


Fig.7

8. Analyze the Frame as shown in Fig. 8 using Moment Distribution method CO4 (20) and also Draw BMD.

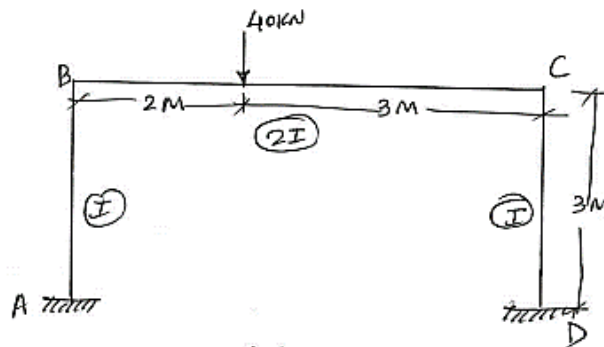


Fig. 8

UNIT - V

9. Analyze the continuous beam shown in Fig. 9 by stiffness matrix method. CO5 (20) Take EI constant throughout.

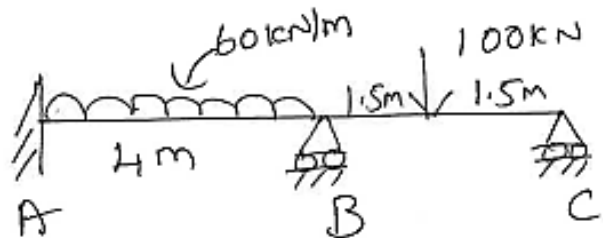


Fig. 9

10. Analyze the given frame shown in Fig.10 using flexibility matrix method. CO5 (20)

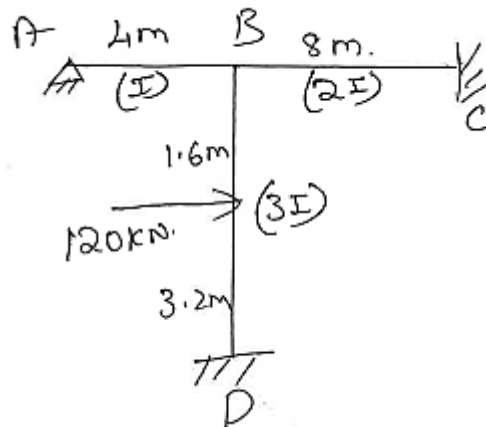


Fig.10
